Organizational Performance Measures for Business Process Management: a Performance Measurement Guideline

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Abstract

This paper focuses on organizational performance measurement. It offers an overview of the literature on organizational performance measures, approaches and frameworks. Analysis indicates that organizational performance measurement is well recognized as an important part of the business process management literature. The purpose of this paper is to propose a performance measurement guideline. Defining an appropriate guideline helps to clarify and systemize this field, but also represents a critical step for business practitioners since it could influence the success of organizational performance measurement system development.

1. Introduction

Business process management (BPM) refers to aligning processes with the organization's strategic goals, designing and implementing process architectures, establishing process measurement systems that align with organizational goals, and educating and organizing managers so that they will manage processes effectively [3; 7; 19; 4]. This term is occasionally used to refer to various automation efforts like Workflow Management (WFM) systems, XML business process languages and ERP solutions, but this approach is too narrow and it does not comprise the entire BPM concept.

BPM enables the design, analysis, optimization, automation and diagnosis of business processes by separating process logic from the applications that run them; managing relationships among process participants; integrating internal and external process resources; and monitoring process performance. Business process management system (BPMS) has built-in metrics that provides process-level instrumentation that can guide process optimization and incremental refinement [24]. According to Khan [14], the measurement and evaluation of the efficiency of business processes is a very important facet of BPMS since it provides real-time feedback about the status of processes and since it measures the time and cost of the processes so that they can be optimized.

It is widely held that performance information enables organizations to gain competitive advantage. Every organization measures, monitors and analyzes its performance. The usual argument for performance measurement tends to rely on a statement such as: "What cannot be measured, cannot be managed". The main reason to implement a measurement system is to increase the overall effectiveness of an organization and its business processes. Thus, organizations are investing large amounts of resources (human and financial) in developing new and efficient performance management systems. Some have very complex and sophisticated performance measurement systems that allow them to track what is happening in real time. Many companies have developed a wide variety of measures which they review periodically. Kueng [15] defines a performance measurement system as an information system which: (1) gathers (through a set of indicators) performance relevant data; (2) compares the current values against historical or planned values, and (3) disseminates the results to the process actors and managers.

However, there is a lack of systematic empirical evidence which has investigated the extent of BPM projects implementation and their effect on

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companies' performance. Although the literature on BPM initiatives includes a rich spectrum of works, there is not enough evidence on their impact on organizational performance. A single definition of organizational performance measures still does not exist. Performance measures have been defined in different ways by different authors. Traditionally, organizational performance has been measured by using financial indicators, but nowadays several non-financial categories are added to performance measurement systems.

The aim of this paper is to:

- present and discuss current literature as to what insight it offers into the use of performance measurement in the area of BPM;
- establish a set of organizational performance measurement elements;
- propose a metrics guide for BPM projects.

In the following section, a number of performance measurement approaches and frameworks is described in order to provide a review of the theoretical literature. The next section presents recommendations for performance measures selection and a business process performance metrics guide proposed by the authors.

2. Organizational Performance Measures, Approaches and Frameworks

Organizational performance comprises the actual output or results of an organization as measured against its intended outputs: goals and objectives. Organizational performance measures allow companies to focus attention on areas that need improvement by assessing how well work is done. With the pressure of global competition, organizational performance measurement has become increasingly necessary for the continued survival of today's companies. Inadequacies in financial performance measures have led to innovations ranging from non-financial indicators of "intangible assets" and "intellectual capital" to "balanced scorecards" of integrated financial and non-financial measures. By supplementing the right set of finnacial measures (e.g. return on investment, return on sales, return on equity, revenue growth, sales by employee, sales to assets, return on assets, cost-income and others) with non-financial data about strategic performance and implementation of strategic plans, companies can communicate objectives and provide incentives for managers to address long-term strategy.

To address this issue, an interdisciplinary review of organizational performance measurement frameworks is espoused in both academic literature and business press [26; 16; 6; 17]. Therefore, there is an extensive amount of literature on performance measures, approaches and frameworks.

Attempts have been made in the past to measure performance based on quantitative financial measures, while less emphasis has been placed on the qualitative components of performance measurement. Hence, Maskell [18] suggests that performance measures should primarily use non-financial performance techniques and change over time as the company needs change. It is also important to involve qualitative indicators, such as customer service and satisfaction, product quality, learning and innovation [13; 20; 21]. In order to investigate the relationship between the implementation of TQM practices and organizational performance in SMEs, Demirbag et.al [5] proposed both financial and non-financial perspectives.

One cannot evaluate organizational performance without taking organizational goals into consideration. The modern business environment demands a multi-goal orientation. According to Waggoner et. al [26] performance measures within an organization can be designed on the basis of 6 different approaches: (1) the engineering approach which measures the input/output ratio; (2) the system approach which sets objectives for each work unit and measures the achievement of these objectives; (3) the management accounting approach measuring the achievement of financial results; (4) the statistical approach which extends the engineering approach by providing empirically tested information about input/output processes; (5) the consumer marketing approach which measures consumer satisfaction and (6) the 'conformance to specifications' approach which advocates the use of a checklist of attributes of a product or service and its service delivery system.

It has become quite obvious that all stakeholders need to be taken into account when assessing modern company's performance. Emerging management paradigms emphasize a stakeholder perspective [1; 2; 9; 10; 23; 22; 25]. When studying the relationship between stakeholder management and a firm's financial performance, Berman et al. [2] found that fostering positive connections with key stakeholders (customers and employees) can help a firm's profitability.

Due to the significance of various stakeholders, organizational performance should not be solely assessed by financial indicators, since they have many limitations and flaws (financial indicators reflect the past decisions and actions; many enhancement cannot be quantified financially; they are inflexible and unified, thus cannot cater to departmental specifics). Further more, focusing solely on financial indicators can cause many problems for a company (managerial focus on short term results for example) [25]. There are several approaches to organizational performance measurement that encompass different stakeholder perspectives. The Balanced Scorecard (BSC) [11; 12; 13; 20] is the most established and most commonly used, but by far not the only one. According to Kueng [15] several process and/or organizational performance measurement approaches have been implemented in the last few years: BSC, self-assessment approach, workflow-based monitoring and statistical process control. Besides these approaches, various tools are used nowadays: Activity-Based Costing systems, the Capability Maturity Model and ISO certification.

The multi-model performance framework (MMPF) model by Weerakoon [27] is also very interesting and has four dimensions including employee motivation, market performance, productivity performance, and societal impact, and covers the satisfaction of various stakeholders such as customers, investors, employees, suppliers and society. According to Tangem [25], more recently developed conceptual framework is the performance prism, which suggests that a performance measurement system should be organized around five distinct but linked perspectives of performance: stakeholder satisfaction, strategies, processes, capabilities and stakeholder contributions. There are many other frameworks that include non-financial perspective i.e. the performance pyramid and the Sink and Tuttle model [25].

3. Performance measurement guideline

In order to achieve business excellence, it is necessary for an organization to develop a system for performance measurement. The initial step to performance system development is to define the metrics guide as a set of recommendations to be followed as the basis for development. This step comprises the identification and definition of performance measures. The second step is to select between various performance measurement frameworks developed in many fields during the last two decades. The final step is to develop a performance measurement system.

Most of the performance measurement systems developed and implemented in organizations are a result of business practices that have been integrated with various performance measurement frameworks. According to Harmon [8] most companies are still experimenting with the specification of process-based performance measures. They rarely have their measures aligned with their strategic goals, but a few companies have begun to explore the integration of frameworks, with their well-defined systems of measures, and the BSC.

The goal of this paper is to clarify the first step of performance measurement system development by making suggestions and recommendations related to the measures of performance measurement system. Since there is no standardized and unified solution, this paper outlines answers to five critical questions in order to help business practitioners to identify and select appropriate metrics.

3.1. Recommendations for Performance Measures

The synthesis of the literature presented in Section 2 is used to develop recommendations for organizational performance measures selection. This set of recommendations describes several types of metrics that have been used effectively in previous BPM projects and other business initiatives. Since the Balanced Scorecard is the most commonly used, it is suggested to use four performance related categories: (1) financial; (2) customer-related; (3) internal business processes-related and (4) learning and growth-related. These strategic perspectives must be elaborated to make them applicable at the tactical level because they act as performance indicator areas. It is also very important to align them to a process view of the organization since they are taken as starting points for the development of performance measurement systems within BPM projects.

The financial performance perspective shows the results of strategic decisions made in other perspectives. Since processes require resources (financial and non-financial) and create added value for customers, they have a significant impact on the financial situation of a company. Efficiently and effectively executed processes are related to the adequate profitability of a company. The customer perspective captures the ability of the organization to provide quality goods and services, the effectiveness of delivery and customers' satisfaction. The internal business process perspective is focused on the analysis of the organization's internal processes. This perspective is very important because it is focused on internal business results that lead to financial success and satisfied customers. The learning and growth perspective explores efficient knowledge, human resources and technology management. Processes will be efficient and effective if adequately skilled and motivated employees supplied with accurate information are driving them. However, if processes are not permanently adapted, process performance will decrease. To avoid it, innovations are needed. Therefore, it is important that each process actor become active in learning and information sharing.

Using these four perspectives, associated business goals and metrics can be defined. Table 1 presents four key performance indicator (KPI) areas and some examples of their business goals. The next step is to define the metrics appropriate for evaluating each performance indicator area, and offer definitions of how each measure can be calculated. Some examples are given in Table 2.

KPI area	Business goals
Financial performance	Revenue growth Net profits Return on assets
Customer related performance	Customer satisfaction Customer oriented solutions Market orientation, development and growth
Internal business processes related performance	Efficient and effective business processes Cost of producing Quality of internal outputs
Learning and growth related performance	Satisfied employees Knowledge base Investment in R&D and innovations

Table 1 KPI areas and their business goals

KPI area and business goals	Metrics and norms
Financial performance: Revenue growth	Revenue from new sources (norm: % of total revenue) Increased market share (norm: % in last three years)
Customer related performance: Customer satisfaction	Increased delivery performance (norm: % of on-time delivery, % goods delivered in 24 hours) Increased customer satisfaction as measured on survey (norm: index)
Internal business processes related performance: Quality of internal outputs	Improved reliability of product delivery (norm: % of in-time delivery with respect to operation schedule) Improved responsiveness (norm: % of production activities cycle time realized according to operation planning)
Learning and growth related performance: Satisfied employees	Decreased staff turnover (norm: 5%) Increased motivation of employees (norm: average number of overtime hours per person)

Table 2 Examples of metrics and norms related to KPI areas and their business goals

If a company has structured its business processes hierarchically, then it becomes necessary to proceed to define appropriate business goals, measures and norms for evaluating each process at each level. Each goal at different level has to be provided with one or more metrics.

3.2. Metrics Guide

In order to clearly identify what should be measured, it is necessary to answer several questions, such as:

- What is the business objective of a BPM project? How is it aligned with the organizational strategy?

Performance measures should be aligned with organizational goals and objectives. Therefore, it is important to focus measures on factors that are important for achieving strategic objectives. The proposition is that any organization should develop a performance measurement system that focuses on strategic aims before embarking on any BPM project.

- Who are the stakeholders? What is the information they need?

A very important step in the measurement process is to identify who will use the measures. This can be business process manager, business process owner, participant of BPM project, funding and approval officials, internal customers, customers and business partners, and other stakeholders. The next step is to identify the stakeholders' most important questions and decisions to define exactly what information they need. Their requirements can be very different, thus measures have to be tailored to each need.

- What will be measured? How will the input data be collected and analyzed?

After identifying the measures for BPM initiative, it is necessary to identify a process for collecting these measures. For system measures, automated data collection systems can be used, such as tools that measure database accesses and "wait times". System-performance logs will also provide valuable system measures. For output and outcome measures, it will be useful to conduct manual counts, estimates and surveys. Interviews, workshops, meetings and brainstorming involving stakeholders can be useful too. Organizational documents contain useful information regarding organizational goals, measures and problems.

- Which approaches, methods and tools will be used?

As described in Section 2, numerous approaches and frameworks using different methods to measure organizational performance have been made. Consequently, different tools supporting these methods have been implemented in the last few years. The question is: "Which approach is appropriate?" Unfortunately, there is no single, simple or universal answer. However, a modern, process-oriented organization should introduce a performance measurement approach supporting a process-oriented view. The second, very important requirement, is to measure organizational performance in two dimensions: quantitatively and qualitatively. Performing measurement is costly and time-consuming. Therefore, it is important to consider the entire set of different measures and then to select the important ones (key performance indicators – KPIs) to use in practice.

- What do the measures tell us?

The measures should be used as a basis to support analysis and decision making. The BPM initiatives measures should be related to (or the same as) existing measures in the organization that are used to monitor the success of business strategy. It is important that executive managers understand that they will get what they have decided to measure. Hence, time and knowledge spent on identifying, selecting and defining appropriate measures aligned with corporate goals, are of high importance and usually well spent.

4. Conclusion

BPM initiatives should be evaluated continually for their progress in achieving organizational performance improvement. Because of the complex and dynamic nature of modern organizations and their environment, BPM initiatives cannot guarantee the successful outcome of plans and goals. However, well-designed performance measures can help managers to review, monitor and evaluate the results of BPM projects. Performance measures support decision making and communication throughout an organization to understand the efficiency results and strategic alignment of BPM projects. Since the value of BPM depends on each organization's goals, this guide is not a set of standard procedures but rather a helpful support to identify and apply appropriate metrics.

The most critical answers of organizational performance measures, frameworks and systems adoption have been recognized and discussed in the paper, though there are still factors to be identified and analyzed through further study. The authors plan to extend this research through the procedure for organizational performance measurement framework selection and organizational performance measurement system development in process-oriented organizations.

References:

[1] Atkinson, A.A., Waterhouse, J.H., Wells, R.B. (1997), "A stakeholder approach to strategic performance measurement", *Sloan Management Review*, Vol. 38, pp. 25-37.

[2] Berman, S.L., Wicks, A.C., Kotha, S., Jones, T.M. (1999), "Does stakeholder orientation matter? The relationship between stakeholder management models and a firm financial performance", *Academy of Management Journal*, Vol. 42 No. 5, pp. 488-506.

[3] Burlton, R.T. (2003), *Business Process Management: Profiting form Processes*, Sams Publishing.

[4] Chaffey, D. and Wood, S. (2005), *Business Information Management: Improving Performance Using Information Systems*, Prentice Hall.

[5] Demirbag, M., Tatoglu, E., Tekinkus, M. and Zaim, S. (2006), "An analysis of the relationship between TQM implementation and organizational performace", *Journal of Manufacturing Technology Management*, Vol. 17, No. 6, pp. 829-847.

[6] Folan, P., Browne, J. (2005), "A review of performance measurement: Towards performance management", *Computers in Industry*, Vol. 56, pp. 663-680.

[7] Greasley, A. and Barlow, S. (1998). "Using simulation modelling for BPR: resource allocation in a police custody process", *International Journal of Operations & Production Management*, No. 18, pp. 978-988.

[8] Harmon, P. (2007), "*Business Process Change: a Guide for Business Managers and BPM and Six Sigma Professionals*", Morgan Kaufmann Publishers.

[9] Harrison, J. S., Freeman, J. S. (1999), "Stakeholders, social responsibility, and performance: empirical

evidence and theoretical perspectives", *Academy of Management Journal*, Vol. 42, pp. 479-485.

[10] Hillman, A.J., Keim, G.D. (2001), "Shareholder value, stakeholder management, and social issues: What's the bottom line?", *Strategic Management Journal*, Vol. 22 No. 2, pp. 125.

[11] Kaplan, R.S., Norton, D.P. (1992), "The Balanced Scorecard – Measures that Drive Performance", *Harvard Business Review*, Vol. 70 No. 1, pp. 71-79.

[12] Kaplan, R.S., Norton, D.P. (1993), "Putting the balanced scorecard to work", *Harvard Business Review*, Vol. 9/10, pp. 134–147.

[13] Kaplan, R.S., Norton, D.P. (1996), "Using the balanced scorecard as strategic management system", *Harvard Business Review*, Vol. 1/2, pp. 75–85.

[14] Khan, R.N. (2003), "*Business Process Management-A Practical Guide*", Meghan-Kiffer Press.

[15] Kueng, P. (2000), "Process performance measurement system: a tool to support process-based organizations", *Total Quality Management*, Vol. 11, No. 1, pp. 67-85.

[16] Kuwaiti, M.E. and Kay, J. (2000), "The role of performance measurement in business process re-engineering", *International Journal of Operations & Production Management*, Vol. 20, No. 12, pp. 1411-1426.

[17] Lin, C.Y., Chen, M.Y. (2007), "Does innovation lead to performance? An empirical study of SMEs in Taiwan", *Management Research News*, Vol. 30, No.2, pp. 115-132.

[18] Maskell, B. (1992), "Performance measurement for world class manufacturing", *Corporate Controller*, January-February 1992, pp. 44-48.

[19] Miers, D. and Harmon, P. (2005), "The 2005 BPM Suites Report", Business Process Trends, http://www.bptrends.com

[20] Neely, A. (2002), "*Business Performance Measurement: Theory and Practice*", Cambridge University Press, Cambridge.

[21] Neely, A., Adams, C. and Kennerley, M. (2002), "*The Performance Prism: The Scorecard for Measuring and Managing Business Success*", Financial Times-Prentice Hall, London.

[22] Riahi-Belkaoui, A. (2003), "Intellectual capital and firm performance of US multinational firms – a study of the resource-based and stakeholder views", *Journal of Intellectual Capital*, Vol. 4 No. 2, pp. 215-226.

[23] Sirgy, J. M. (2002), "Measuring corporate performance by building on the stakeholder model of business ethics", *Journal of Business Ethics*, Vol. 35 No. 3, pp. 143-162.

[24] Smith, H. and Fingar, P. (2003), "*Business Process Management: the Third Wave*", Meghan-Kiffer Press.

[25] Tangem, S. (2004), "Performance measurement: from philosophy to practice", *International Journal of Productivity and Performance Management*, Vol. 53 No. 8, pp. 726-737.

[26] Waggoner, D.B., Neely, A.D. and Kennerley, M.P. (1999), "The forces that shape organizational performance measurement systems: An interdisciplinary review", *International Journal of Production and Economics*, Vol. 60-61, pp. 53-66.

[27] Weerakoon, T.S., (1996), "Organizational performance- a stakeholder concept", International Research Conference on Quality Management Proceeding, pp. 80-90.